Big Data

Management and Analytics

TU Clausthal, Institut für Informatik

Summer Semester 2026

Due Date: 3 May 2026, 16:00

Instructions — Please Read Carefully

In this assignment, we practice frequent itemsets and association rules. For Task 1, you will be required to practice these concepts on the provided dataset (no programming), and for Task 2, you will be required to use the “E-Commerce”-DB in Neo4j.

Cypher Reference: https://neo4j.com/docs/cypher-refcard/current/

Include all essential ideas and steps necessary to derive a solution. Write your solution in an understandable form or draw the structure with an appropriate tool. If there is missing information, make an assumption and state it explicitly — it will hold for the entire assignment unless declared otherwise.

Assignment Rules

- The assignment has a total of **10 marks**
- Work in your assigned teams of **up to 2 members**
- Submit your solutions **via Moodle**
- Solutions submitted after the deadline will automatically receive **0 marks**
- Copying solutions from or sharing with other students will lead to **exclusion from the course**
- The use of language software is **not permitted**; each group member must understand every part of the solution
- Document your solutions with **screenshots** and attach a **TXT file** with your code/queries
- If anything is unclear, state your assumptions explicitly to make your solution understandable

Task 1 5 Marks

This task focuses on **frequent itemsets** and **association rules**. Consider the following table with baskets of a hobby/electronics shop, together with the respective customer ID and transaction ID.

CustomerID	TransactionID	Basket
1	6104	{Laptop, Wireless Mouse, Sci-Fi Book}
2	7281	{Spices Pack, Classic Poster, Rubik's Cube}
3	3469	{Box of Chocolates, Horror Novel}
4	9052	{Classic Poster, Spices Pack, Hammer}
5	4817	{Spices Pack, Rubik's Cube}
6	2635	{Sci-Fi Book, Box of Chocolates, Laptop}
7	8196	{Soap Bar, Spices Pack, Anime Figurine}
8	5370	{Horror Novel, Hammer, Classic Poster}
9	6748	{Anime Figurine, Laptop}
10	3921	{Replica Sword, Soap Bar, Comic Book}
11	7564	{Laptop, Anime Figurine, Language Textbook}
12	1289	{Classic Poster, Language Textbook, Comic Book}
13	9437	{Wireless Mouse, Sci-Fi Book, Laptop}

(a) Compute the support for the itemsets {Classic Poster}, {Rubik's Cube, Spices Pack}, and {Box of Chocolates, Sci-Fi Book, Horror Novel}.

(b) Compute the confidence for the following association rules:

- {Sci-Fi Book, Wireless Mouse} $\rightarrow$ {Laptop}
- {Spices Pack, Rubik's Cube} $\rightarrow$ {Classic Poster}
- {Soap Bar} $\rightarrow$ {Language Textbook}

Explain why the results for the last two rules are different.

(c) Assume we have the support threshold $s = 3$. List all frequent itemsets.

(d) Now assume we have the support threshold $s = 4$. Apply the **A-priori algorithm** on the given set of baskets. *Show the frequent itemsets for each iteration.*

Task 2 — 5 Marks

This task focuses on discovering **product associations** in the **E-Commerce** database using **Cypher**. Write one Cypher query for each subtask and briefly explain what your query does.

(a) Find all pairs of products that were ordered together in the same order.
Return the two product names and the number of times they appeared together, ordered by co-occurrence count in descending order.

(b) Calculate the support for each product, defined as the number of orders containing that product divided by the total number of orders in the database.
Return the product name and its support value, ordered by support in descending order.

(c) Find all pairs of products that were ordered together in the same order with a minimum support of 0.0002.
Return the two product names and their support value.

(d) For each pair of products $(A \to B)$ that appear together in orders, calculate the confidence of the rule, defined as:

$$\text{confidence}(A \to B) = \frac{\text{orders containing both } A \text{ and } B}{\text{orders containing } A}$$

Return product A, product B, and the confidence value. Only return rules with a confidence above 0.15, ordered by confidence in descending order.

(e) For each product pair $(A \to B)$, calculate the lift of the association rule, defined as:

$$\text{lift}(A \to B) = \frac{\text{confidence}(A \to B)}{\text{support}(B)}$$

A lift greater than 1 means the products are positively correlated. Return product A, product B, confidence, support of B, and lift. Only return rules where lift > 1, ordered by lift in descending order.

Deliverables: Submit a `.txt` file containing all Cypher queries and screenshots showing the execution results in Neo4j.

End of Assignment